Time left 2:29:09

Advancement of operating system is closely tied to computer architectures, therefore, has passed through different generations. Which one of the following generation and advancement is wrongly coupled?

- a. Second generation ->multiprogramming
- b. Fourth generation -> real time systems
- c. First generation-> single user
- d. Third generation -> batch system

Which one of the following is asynchronous in message communication?

- a. Non-block receive
- b. Blocking receive
- c. Blocking send
- d. Direct message

Which one of the following is not correct about transaction failure?

- a. Transactions fail if logical errors are detected
- b. Catastrophes can be cause of transaction failure
- c. Concurrency control enforcement avoids transaction failure
- d. Transactions fail if disk blocks lose their data

Which one of data updates best describes the following statement? "All modified data items in the cache is written either after transaction ends its execution or after a pre-determined number of transactions have concluded their execution"

- a. Immediate update
- b. In-place update
- c. Shadow update
- d. Deferred update

Which one of the following is true about functions in JavaScript?

- a. Function names can contain spaces
- b. Function names are not case-sensitive
- c. There is no limit to the number of function parameters that a function may contain
- d. Function names can begin with digits

If you plan a software test to be done by clients to demonstrate that the system satisfies their requirements, which phase of software testing is appropriate?

- a. Acceptance testing
- b. Integration testing
- c. System testing
- d. Unit testing

Assume you have a weighted graph G with a set of vertices V and set of edges E. Which one of the following problem is an optimization problem on the graph?

- a. Traverse all vertices in graph G
- b. Is vertex pϵV reachable from vertex qϵV in graph G
- c. Find minimum spanning tree from graph G
- d. How many cycles are there in graph G

What is the time complexity order of binary searching algorithm?

- a. $O\left(n^3\right)$
- b. $O(n)$
- c. $O\left(\log_2 n\right)$
- d. $O(1)$

Which statement is false about programming language generations?

- a. Assembly language is machine independent
- b. Machine language can be executed without translation
- c. Assembly language should be translated to machine language before execution
- d. Fifth generation programming languages need very powerful hardware and software

Suppose you have developed a self-driving car equipped with relevant sensors and camera that helps it to see the road and drive sensing traffic lights, pedestrians and signs. Its sensors are able to detect the environment but not all the aspects. Hence, in which environment is your car acting?

- a. Stochastic
- b. Discrete
- c. Fully observable
- d. Partially observable

Which one of the following is not correct about threads?

- a. Treads are the entities schedule for execution on the CPU
- b. It is a group of processes
- c. Treads have program counters
- d. Treads have registers to hold its working memory

If you want to allow subclasses to access data fields or methods defined in the super-class, but not to allow non-subclasses to access these data fields and methods, which access modifier is appropriate?

- a. protected
- b. private
- c. default
- d. public

Clear my choice

An interface that has multiple lines connecting input/output module and peripheral device and assures multiple bits to be transferred at the same time is:

- a. Bus interface
- b. Parallel interface
- c. One line interface
- d. Serial interface

Look at the following PHP line: $list = array(20, 12, 23, 19, 18); What is the index of data value 19?

- a. 2
- b. 0
- c. 3
- d. 1

Which one of the following is correct for the statement below? "A problem that occurs when one transaction updates a database item and then the transaction fails for some reason and the updated item is accessed by another transaction before it is changed back to its original value"

- a. It is a temporary read problem
- b. It is dirty read
- c. It is an incorrect summary problem
- d. It is a lost update problem

What is the time complexity order of Quick sort algorithm?

- a. $O(n)$
- b. $O\left(n^2\right)$
- c. $O\left(2^n\right)$
- d. $O\left(n \log_2 n\right)$

Which one of the following is not a single level ordered index?

- a. Secondary index
- b. Primary index
- c. Clustering index
- d. Multilevel index

Which branch of study deals with whether a problem can be solved at all or not, regardless of the resources required?

- a. Computability theory
- b. Automata theory
- c. Set theory
- d. Complexity theory

Which one of the following is correct way creating String object?

- a. String m = 'Computer Science';
- b. String m = new String('Technology');
- c. String m = 'H', 'e', 'l', 'l', 'o';
- d. String m = new String("Exit Exam");

Which quantifier can be used for some portion of the universe?

- a. ∀
- b. →
- c. ∃
- d. ^

Clear my choice

Which one of the following is not correct about POP3 and SMPT?

- a. A sender can send email using SMTP
- b. Both are application layer protocols
- c. A sender can receive emails using POP3
- d. Both are network layer protocols

A university owns a number of private local area and wide area networks, which are designed for an access by its academic staff, students and administrative workers. Which term best describes the University's network?

- a. Local Area Network
- b. The Internet
- c. Wide Area Network
- d. Metropolitan Area Network

Among the following, which one best describes a database management system (DBMS)?

- a. It is responsible for creating and maintaining a database
- b. It can help users to manage computer's activity
- c. It can assist process images
- d. It can assist you to process document

Let a Grammar $\mathbf{G} = (\{\mathrm{S}\}, \{\mathrm{a}, \mathrm{b}\}, \mathrm{S}, \mathrm{P})$, where $\{S\}$ is set of variables, $\{a, b\}$ is set of terminal symbols, S is start variable and P is set of productions. Then which one of the following is not an element of the language $(\mathbf{L})$ (L) which is generated from G. Given:

$\mathbf{P : S \rightarrow aSb,}$
$\mathbf{S \rightarrow \lambda}$

- a. aabb
- b. ab
- c. $\boldsymbol{\lambda}$
- d. a

At which layer of the OSI model devices such as bridges, switches and Network Interface cards are used?

- a. Physical layer
- b. Application layer
- c. Network layer
- d. Data link layer

Consider the following fragment of C++ code:

```
const int x = 8;
const int y = 16;
if(x > y)
{
        int z = x + y;
}
```

For codes like the above one, which technique of code optimization is appropriate?

- a. Code motion
- b. Dead code elimination
- c. Compile time evaluation
- d. Arithmetic simplification

Assume you make modification to your software. Then you want to make sure that the modification has not had introduced new errors. Which software testing level is required?

- a. Regression testing
- b. Static testing
- c. Beta testing
- d. Dynamic testing

Suppose you are administering a network and would like to know whether a computer in the upstairs is connected to the network or not. Therefore, which command would you use?

- a. nslookup
- b. ping
- c. DHCP discover
- d. traceroute

A computer expert would like to know the number of processes that are completed per hour to assess the performance of a machine. Therefore, which one of the following best describes the expert's assessment?

- a. Waiting time
- b. System throughput
- c. Turnaround time
- d. Response time

Which one of the following is false about arrays in C++?

- a. We use [ ] square bracket at the time of array declaration
- b. Size of an array should be constant at the time of array declaration
- c. An array is a collection of similar data objects
- d. We can access elements of arrays without using index number

Which asymptotic notation describes average case analysis?

- a. $\mathbf{f(n) = \Omega(\text{ Big omega })}$
- b. $\mathbf{f(n) = \text{ Big-Oh}}$
- c. $\mathbf{f(n) = \text{ little-}\ \omega(\text{ little omega })}$
- d. $\mathbf{f(n) = \theta\ (\text{thetha})}$

What is the output of the following fragment C++ code?

```
for(int n = 1; n < =  18; n = n + 2)
{
        if(n % 7 ! = 0)
            cout<<n<<"   ";
        else
            break;
}
```

- a. 3 5 7
- b. 1 3 5 7 9 11 13 15 17
- c. 1 3 5
- d. 1 2 3 4 5 6

Which one of the following is a two-level cache having an internal and external caches?

- a. Multilevel cache
- b. Single level cache
- c. Split cache
- d. Unified cache

Given the schemas:BOOK (Book_id, Title, Publisher_name) BOOK_AUTHOR (Book_id, Author_name) PUBLISHER (Name, Address, Phone) Which one of the following SQL is correct to delete the publisher named as 'XWZ' from the PUBLISHER table

- a. DELETE FROM BOOK_AUTHOR WHERE Address = 'XWZ'
- b. DELETE FROM PUBLISHER WHERE Name = 'XWZ'
- c. DELETE FROM PUBLISHER WHERE Address = 'XWZ'
- d. DELETE FROM BOOK WHERE Name = 'XWZ'

Which one of the following is different from the other based on its role in computer network?

- a. Routers
- b. Network printer
- c. Wireless access point
- d. Hubs

_____ is a network architecture characteristics that states a network can expand quickly to address needs of new users and application without having impact on the services delivered to existing users.

- a. Security
- b. Quality of service
- c. Fault tolerance
- d. Scalability

Which one of the following is a step in problem solving that lists the actions and states that the agent has to consider provided a goal?

- a. Problem formulation
- b. Execute
- c. Search
- d. Goal formulation

Which network type is the largest as compared to the rest?

- a. Wide Area Network
- b. Metropolitan Area Network
- c. The Internet
- d. Local Area Network

Which one of the following is valid statement regarding PHP variables?

- a. $test = 9.5
- b. $test = "This is 'Computer science' ";
- c. $test = "Computer;
- d. test = 5;

Which one of the following is a measure taken, to ensure information and information systems' availability, integrity, authentication and confidentiality?

- a. Wiretapping
- b. Information security
- c. Information assurance
- d. Interception

Which one of the following phases is done before semantic analysis phase of compiling process?

- a. Syntax analysis
- b. Intermediate code generation
- c. Code generation
- d. Code optimization

Which one of the following lists of environments is the hardest for an agent?

- a. Deterministic, fully observable, static
- b. Fully observable, sequential, deterministic
- c. Partially observable, static, continuous
- d. Sequential, non-deterministic, dynamic

Assume you have a universe of Key U. Which one is false statement regarding Direct Address table and Hash Table data structures?

- a. If universe U is very large, Hash table is better than Direct Address table
- b. Direct Address table doesn't use hash function to map keys
- c. Hash table allocates one separate memory slot for each key in the universe U
- d. There is one to one correspondence between keys in the universe U and memory slots in the Direct Address table

Which one of the following is a security service enforced to protect discloser of information, which can be stored in file or being on transmission, from unauthorized entities?

- a. Authentication
- b. Integrity
- c. Availability
- d. Confidentiality

Which one of the following is correct for the statement below? When two transactions that access the same database items have their operations interleaved in a way that makes the value of some database item incorrect"

- a. It is a temporary read problem
- b. It is a dirty read problem
- c. It is an incorrect summary problem
- d. It is a lost update problem

Which deadlock prevention mechanisms don't require timestamp?

- a. Wait-wait
- b. No-wait
- c. Wound-wait
- d. Wait-die

A database management system (DBMS) has_____ to control locks.

- a. Lock manager
- b. Lock table
- c. Query optimizer
- d. Query processor

What does the program data independence in the database approach entails for?

- a. The application may be affected when data changes
- b. Data and applications are defined separately
- c. Programs can create redundant data independently
- d. Both program and data are defined together

Which one of the following is evaluating the degrees of success of an agent?

- a. Performance measure
- b. Action
- c. Knowledge
- d. Perception

In a tree data structure, if a node has no parent node, then the node is _____.

- a. Parent node
- b. Root node
- c. External node
- d. Internal node

Which one of the following is not correct about computer architecture and organization?

- a. An architecture can remain for long time but its organization can change
- b. Manufacturers offer computer with same architecture but different organization
- c. An architecture can be used for long time encompassing different computer models
- d. An organization can last for long time as an architecture does

Which phase of compiling process is considered as the final phase in the process?

- a. Intermediate code generation
- b. Semantic analysis
- c. Code optimization
- d. Code generation

Which statement is false about universal hashing technique?

- a. In universal hashing a hash function is selected randomly from set of hash functions for each key
- b. In universal hashing a hash function is selected from set of hash functions independent of keys.
- c. Single fixed hash function technique is better than universal hashing technique in terms of collision reduction
- d. Universal hashing reduces the chance of data collision

Which one of the following is responsible for separating some machines from the rest to shield them from possible attacks?

- a. Network operating system
- b. Firewall
- c. Router
- d. Switch

Which type of variable can be accessed through out our C++ program scope?

- a. Static
- b. Automatic
- c. Local
- d. Global

Which type of Turing machine has two tapes, one tape read-only and the other read-write tape?

- a. Multi-head Turing machine
- b. Off-line Turing machine
- c. Non-deterministic Turing machine
- d. Multi-dimensional Turing machine

Which one of the following is not correct about packet and circuit switching?

- a. In packets switching messages are sent in small blocks
- b. In Packet switching and end-to-end connection has to be established
- c. In circuit switching a channel is dedicatedly used
- d. Packet switching is more efficient than circuit switching

Which one of the following best explains attackers who are motivated by financial gain and perform sabotage, an IP theft and secrets of a rival company?

- a. Stealer
- b. Competitor
- c. Hacktivist
- d. Activist

_____ is set of attributes that can be seen by a program and can have direct impact on the logical execution of a program.

- a. Arrangement
- b. Organization
- c. Flow control
- d. Architecture

Which one of the following computers can be most secured compared to the rest?

- a. A computer connected to the Internet with the latest firewall system
- b. A computer connected to the Internet having strong intrusion detection
- c. A computer that is not connected to the Internet
- d. A computer connected to the Internet and the latest anti-virus installed on it

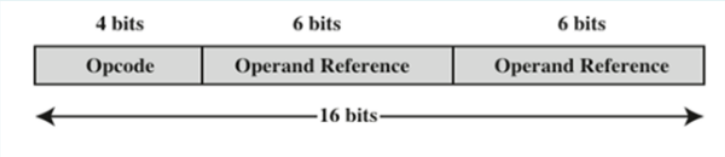
_____ is the whole thing that an agent has observed so far concerning the existing scenario in the environment.

- a. Performance measure
- b. Perception
- c. Knowledge
- d. Action

Which layer uses port number to identify applications?

- a. Transport layer
- b. Physical layer
- c. Network layer
- d. Application layer

What is the purpose of an opcode that is as indicated in the figure below?



- a. It holds the destination address of an operand
- b. It determines the operation to be done
- c. It holds the source address of an operand
- d. It holds the next instruction reference

Assume there is a line of patients waiting for a service from a clinic. Seriously sick patients should get service first. Which data structure is the most appropriate for the given type of real world problem?

- a. Tree
- b. Queue
- c. Stack
- d. Priority Queue

Size of large data files can be reduced to save storage disk space. Which algorithm is appropriate to reduce size of large files?

- a. Heap sort algorithm
- b. Prim's algorithm
- c. Merge sort algorithm
- d. Huffman encoding algorithm

After deep analysis if your software requires major change to the architecture of the system, which strategy of software change should be applied?

- a. Preventative maintenance
- b. Architectural transformation
- c. Software reengineering
- d. Software maintenance

Which one of the following is not correct for the figure shown in Question 27?

- a. P2 cannot be executed until P1 finishes
- b. P1 is waiting for a resource held by P2
- c. P3 cannot be executed until P4 finishes
- d. P0 is waiting for a resource held by P1

Which one of the following is the smallest heading tag?

- a. <H6>
- b. <H3>
- c. <H4>
- d. <H1>

Which one of the following is false about String in Java?

- a. In Java a string is treated as object
- b. We can create a String object using String literal
- c. We can create a String object using array of characters
- d. Content of a string can be changed once the string is created

Which one of the following is not delimiter of PHP code?

- a. <? ..... ?>
- b. <caption> ..... </caption>
- c. <script language="PHP"> ..... </script>
- d. <?php ..... ?>

Clear my choice

Which one of the following is not true about fan trap in Entity Relationship Diagram?

- a. It occurs when there is no pathway between related entities
- b. Highly likely exit when more than two one-to-many relationships fan out from an entity
- c. When there is ambiguous pathway between some entity occurrences in related entity types
- d. It can be resolved by readjusting the model so that no one-to-many relationships fanning out from an entity

Suppose you have received a message from your friend but the message is scrambled for the sake of security. Your friend has used a program to encode messages into scrambled text. In order to read the message which program would you use?

- a. Encryption program
- b. Ciphertext
- c. Secret key
- d. Decryption program

What is the output of the following Java fragment code?

```
int n = 6, m = 15, p = 3;
n -= m;
n *= p;
System.out.print(n);
```

- a. 27
- b. 3
- c. 24
- d. -27

Which concept refers to extending specialized classes from generalized classes in object-oriented programming?

- a. Inheritance
- b. Overridding
- c. Overloading
- d. Data field Encapsulation

A process is terminated by the operating system due to a reference to non-existing memory. Which one of the following best explains the process termination?

- a. Normal exit
- b. Error exit
- c. Terminated by another process
- d. Fatal exit

For an input graph, Depth First Traversal algorithm uses _____ data structure to traverse.

- a. Priority queue
- b. Tree
- c. Queue
- d. Stack

Which one of the following is true about java and javascript?

- a. Both are object oriented languages
- b. Javascript is compiled and interpreted in source code form
- c. Javascript can be used in extremely sophisticated applications
- d. There are more features in javascript than java

Which one of the following does not show non-functional requirement of software systems?

- a. Response time
- b. Memory requirement
- c. Reliability
- d. Displaying information

Which one of the following is false about project planning stage?

- a. Risk analysis is done during project planning
- b. Preparing time schedule is done during project planning
- c. Cost estimation is done during project planning
- d. Project planning is a one-time task in software development life cycle

Which one of the following is true about frames in HTML?

- a. Frames are not difficult to handle for search engines
- b. Load on the server is not affected, if there are a large number of frames in a page
- c. All browsers support frames
- d. Frames allow parts of the page to remain stationary while other parts scroll

Clear my choice

58. Which network security issue shows the level of weakness in networks and devices?

- a. Attack
- b. Availability
- c. Vulnerability
- d. Treat

Which statement is true about pop() operation of Stack data structures?

- a. pop() returns the end/top element but it doesn't delete that element
- b. pop() removes the end/top element from stack
- c. pop() inserts new element at the middle of stack
- d. pop() inserts new element at the beginning/bottom of stack

Which asymptotic notation describes worst case analysis?

- a. $\mathrm{f(n) = \ little\text{-}\ \omega(\ little\ omega\ )}$
- b. $\mathrm{f(n) = \theta\ (thetha)}$
- c. $\mathrm{f(n) = \ little\text{-}oh}$
- d. $\mathrm{f(n) = \Omega(\ Big\ omega\ )}$

Which one of the following is correct about human role in system administration?

- a. Ethics is not a requirement to manage computers
- b. It requires patience, understanding and knowledge
- c. It doesn't need organizational skills
- d. System administrator is not expected to have skills and confidence

What will be the output of the following JavaScript statement? Math.floor(5.9)

- a. 9
- b. 11.8
- c. 6
- d. 5

In which situation that using database management system (DBMS) becomes costly and not recommended to use it?

- a. In embedded systems having too small storage space
- b. In situation where more than one users should access the database
- c. To deny access to unauthorized users
- d. To control redundancy

Which one of the following algorithms can be used to extract Minimum Spanning Tree from an input graph?

- a. Prim's algorithm
- b. Huffman encoding algorithm
- c. Dijkstra's algorithm
- d. Merge sort algorithm

Which one of the following is not an example of scripting languages?

- a. Python
- b. Perl
- c. PHP
- d. C++

What is the output of the following C++ fragment code?

```
int  a = 6, b = 8;
int  x = 2, y = 4;
int  c = (x > y ? (a--, x) : (b--, y));
cout<<"a= "<<a;
cout<<"      b= "<<b;
cout<<"      c= "<<c;
```

- a. a = 5 b = 7 c = 2
- b. a = 5 b = 8 c = 4
- c. a = 6 b = 8 c = 4
- d. a = 6 b = 7 c = 4

Which phase of software development process refers to modification of software to reflect changing customer and market requirements?

- a. Validation
- b. Evolution
- c. Development
- d. Specification

Which one of the following registers hold the address of the next pair of instructions to be fetched from memory?

- a. Memory address register
- b. Memory buffer register
- c. Program counter
- d. Instruction buffer register

What is the output of the following C++ fragment code?

```
int  a, b;
a= 13;
b = 9;
while( (20  && 0) && (a > b) )
{
      cout<<"Plants are our life";
}
```

- a. It displays message "Plants are our life" infinitely
- b. No message will be displayed
- c. It displays message "Plants are our life" 20 times
- d. It displays message "Plants are Our Life" 4 times

Which one of the following is used for the purpose of syntax analysis?

- a. Loader
- b. Linker
- c. Parser
- d. Scanner

Given input Alphabet $\Sigma = \{a, b, c, d\}$ and empty string λ. Then, which one of the following is true about Σ^*?

- a. $\Sigma^* = \Sigma^+$
- b. $\Sigma^* = \Sigma^+ U \{\lambda\}$
- c. $\Sigma^* = \Sigma^1 \mathrm{U} \Sigma^2 \mathrm{U} \Sigma^3 \mathrm{U} \Sigma^4$
- d. $\Sigma^* = \{\lambda\}$

Given input Alphabet $\Sigma = \{a, b, c, d\}$ and empty string λ. Then, which one of the following is equivalent to Σ^0??

- a. Σ
- b. Σ^+
- c. $\{\lambda\}$
- d. Σ^*

Which one is a method used to obtain the meaning of encrypted information without access to secret key?

- a. Cryptography
- b. Crypto-analysis
- c. Decryption
- d. Encryption

Linked list data structure is _____ type of data structure.

- a. Non-linear data structure
- b. Fixed size data structure
- c. Static data structure
- d. Dynamic data structure

What is the output of the following Java fragment code?

```
int[] list = new int[4];
int sum = 0;
for(int i = 0; i < list.length; i++)
{
         list[i] = i*3;
         sum += list[i];
}
System.out.print(sum);
```

- a. 18
- b. 6
- c. 3
- d. 36

What will be the output of the following JavaScript statement? Math.sqrt(36)

- a. 6
- b. 30
- c. 1
- d. 36